FARM MANAGEMENT SYSTEM – PROJECT DOCUMENTATION

Salesforce Based Application

Submitted by

SUGANTHI S

STELLAJOYS

SUBITHA G

SUJITHA R

SUJITHRA K

Department of Computer Applications

1. Abstract

The Farm Management System (FMS) is a cloud-based application developed using the Salesforce platform to manage agricultural operations efficiently. The system allows farmers, farm managers, and administrators to store and manage information related to farms, crops, farmers, and buyers.

The application provides a centralized platform where users can enter farm details, track crop production, manage farmer information, and monitor buyer transactions. The system also includes reports and dashboards that provide real-time insights into farm productivity and sales performance.

By digitizing agricultural data management, the system reduces manual paperwork, improves accuracy, and enables better decision-making.

2.Introduction

Agriculture is an important sector that contributes significantly to the economy. Many farmers still maintain records manually in notebooks or spreadsheets. This traditional method often results in data loss, errors, and inefficient farm management.

The Farm Management System provides a digital solution for managing agricultural information. The system is developed using the Salesforce platform and enables users to manage farm records, crop details, farmer information, and buyer transactions in a centralized environment.

3. Problem Statement

In many agricultural systems, farm records are maintained manually using registers or spreadsheets. This creates several problems such as:

• Data inconsistency
• Difficulty in tracking crop production
• Delayed reporting
• Data loss due to manual storage
• Lack of centralized information

4. Objectives of the Project

The main objectives of the Farm Management System are:

• Digitize farm data management

• Improve data accuracy and reliability

• Reduce manual paperwork

• Enable real-time monitoring of farm activities

• Provide analytical insights through reports and dashboards

• Improve overall farm productivity

5. System Architecture

The system architecture consists of four layers.

Presentation Layer

Salesforce Lightning User Interface used for interacting with the system.

Application Layer

Handles validation rules, automation, and business logic.

Data Layer

Stores all farm-related data in Salesforce custom objects.

Analytics Layer

Generates reports and dashboards for farm data analysis.

6. Technology Stack

Layer	Technology
• Frontend	Salesforce Lightning UI
• Backend	Salesforce Platform
• Database	Salesforce Cloud
• Automation	Validation Rules
• Analytics	Report& Dashboards

7. Modules of the System

The system consists of four major modules.

Farm Module

This module stores information about farms such as farm type, soil quality, irrigation type, location, and size.

Crop Module

This module stores crop details including crop name, planting date, harvest date, yield, and harvested area.

Farmer Module

This module manages farmer information such as name, contact number, state, district, and village.

Buyer Module

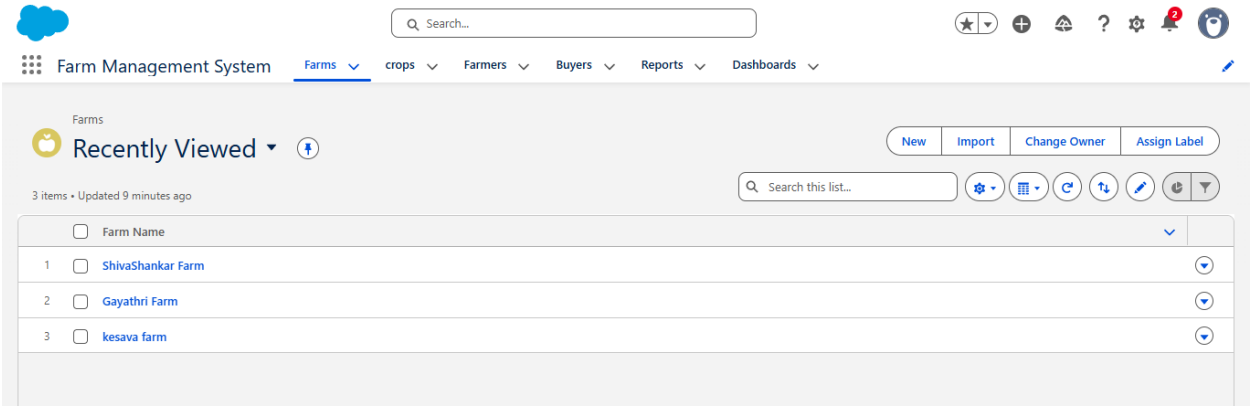
This module stores buyer details including contact number, email address, buyer type, and address.

8. Implementation Steps

The implementation of the Farm Management System involves the following steps:

1. Create Salesforce Developer Account
2. Create Custom Objects (Farm, Crop, Farmer, Buyer)
3. Create Fields for each object
4. Define Relationships between objects
5. Create Tabs for objects
6. Create Lightning Application
7. Configure Profiles and Roles
8. Create Users
9. Implement Sharing Rules

10. Generate Reports and Dashboards



9.Farms

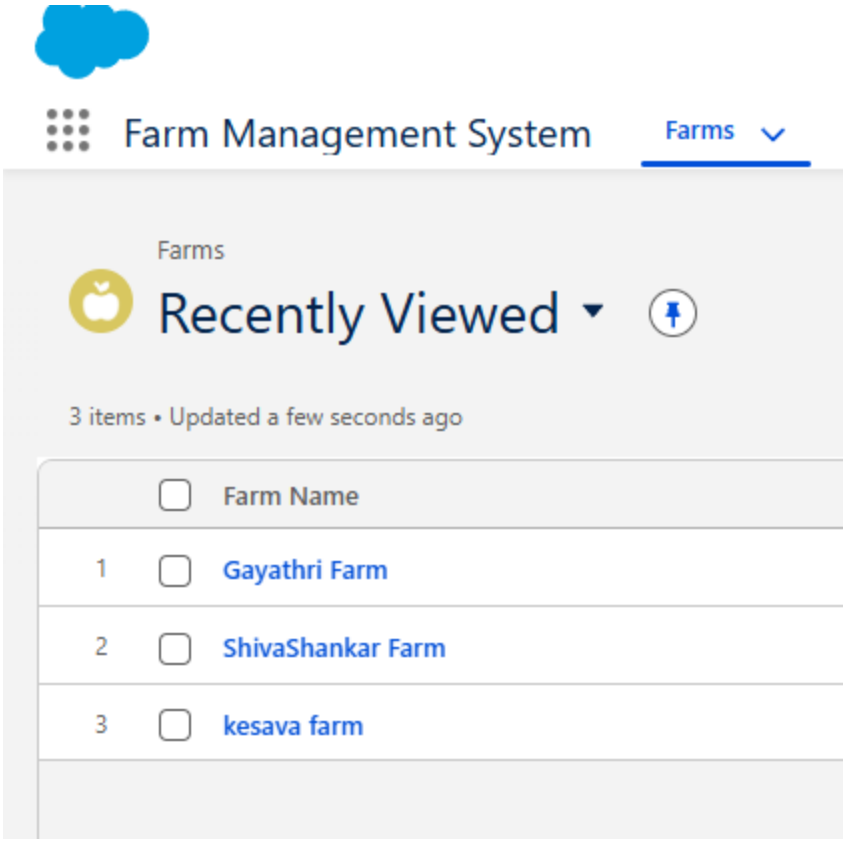
The **Farm module** stores all the details related to agricultural farms. It helps the system maintain information about farm location, soil type, irrigation method, and farm size. This module acts as the main data source for crop production and farm management activities.

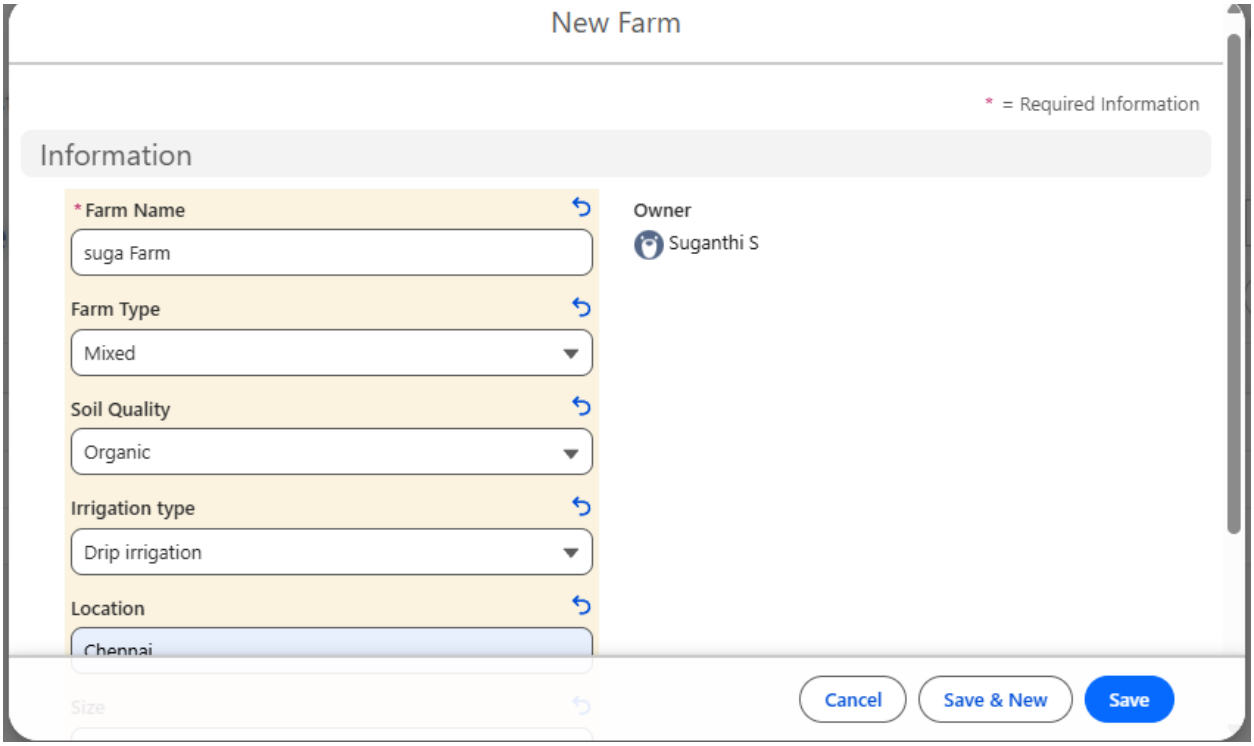
The Farm module allows administrators and farm managers to track different farms and monitor agricultural activities effectively.

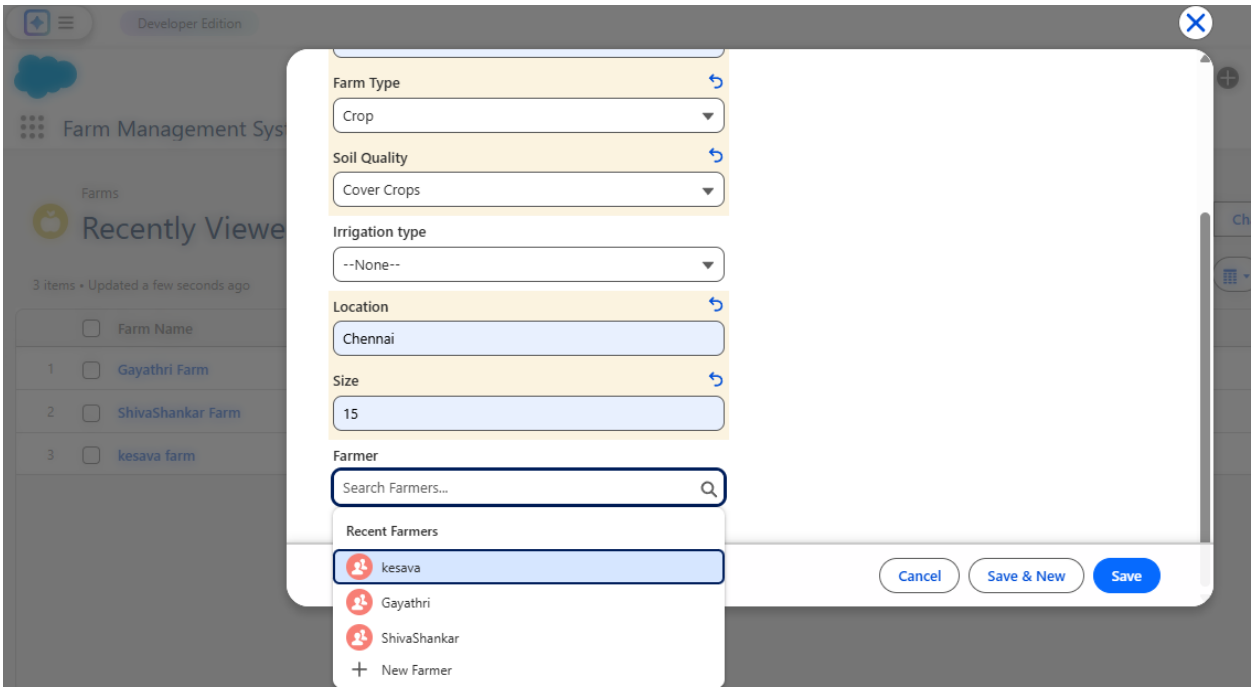
Key Fields in Farm Object

- Farm Name
- Farm Location
- Farm Size (in acres)
- Soil Type
- Irrigation Type
- Farm Owner

Screenshots:


Farm Management System
Farms
Farms
Recently Viewed
3 items • Updated a few seconds ago
Farm Name
1
Gayathri Farm
2
ShivaShankar Farm
3
kesava farm


New Farm
* = Required Information
Information
* Farm Name
suga Farm
Farm Type
Mixed
Soil Quality
Organic
Irrigation type
Drip irrigation
Location
Chennai
Owner
Suganthi S
Cancel
Save & New
Save



10.Crops:

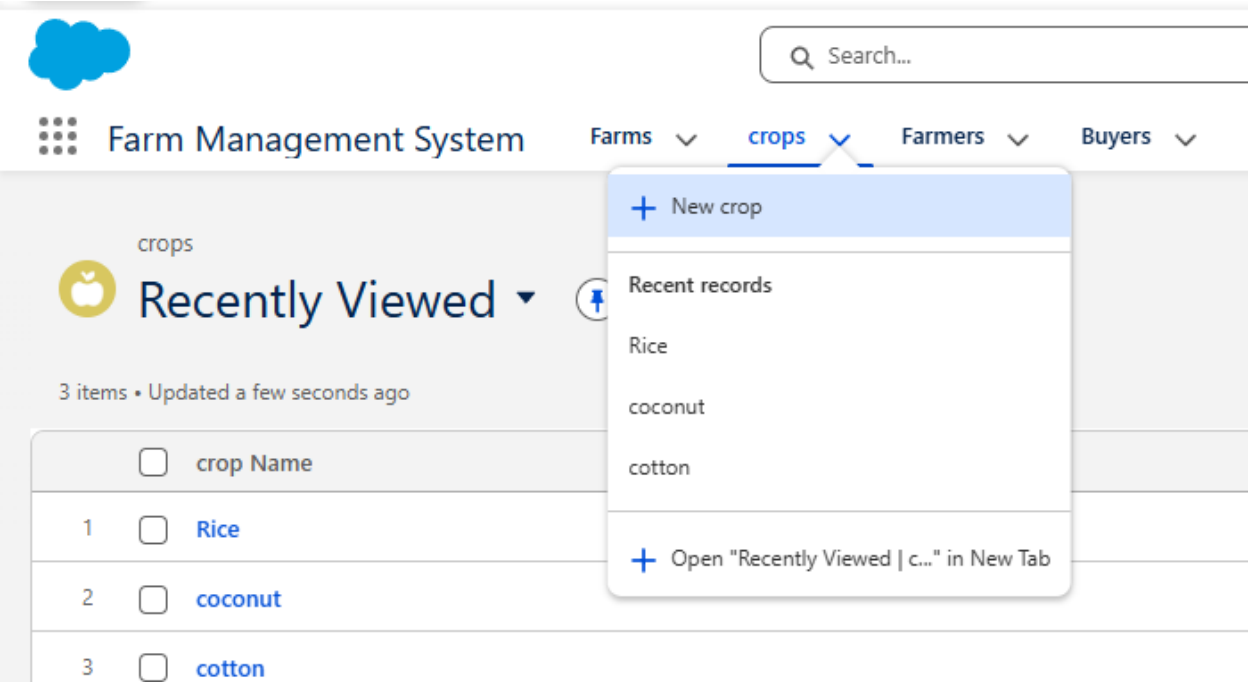
The **Crop module** stores information related to crops cultivated on farms. It records crop details such as crop type, planting date, harvest date, and crop yield.

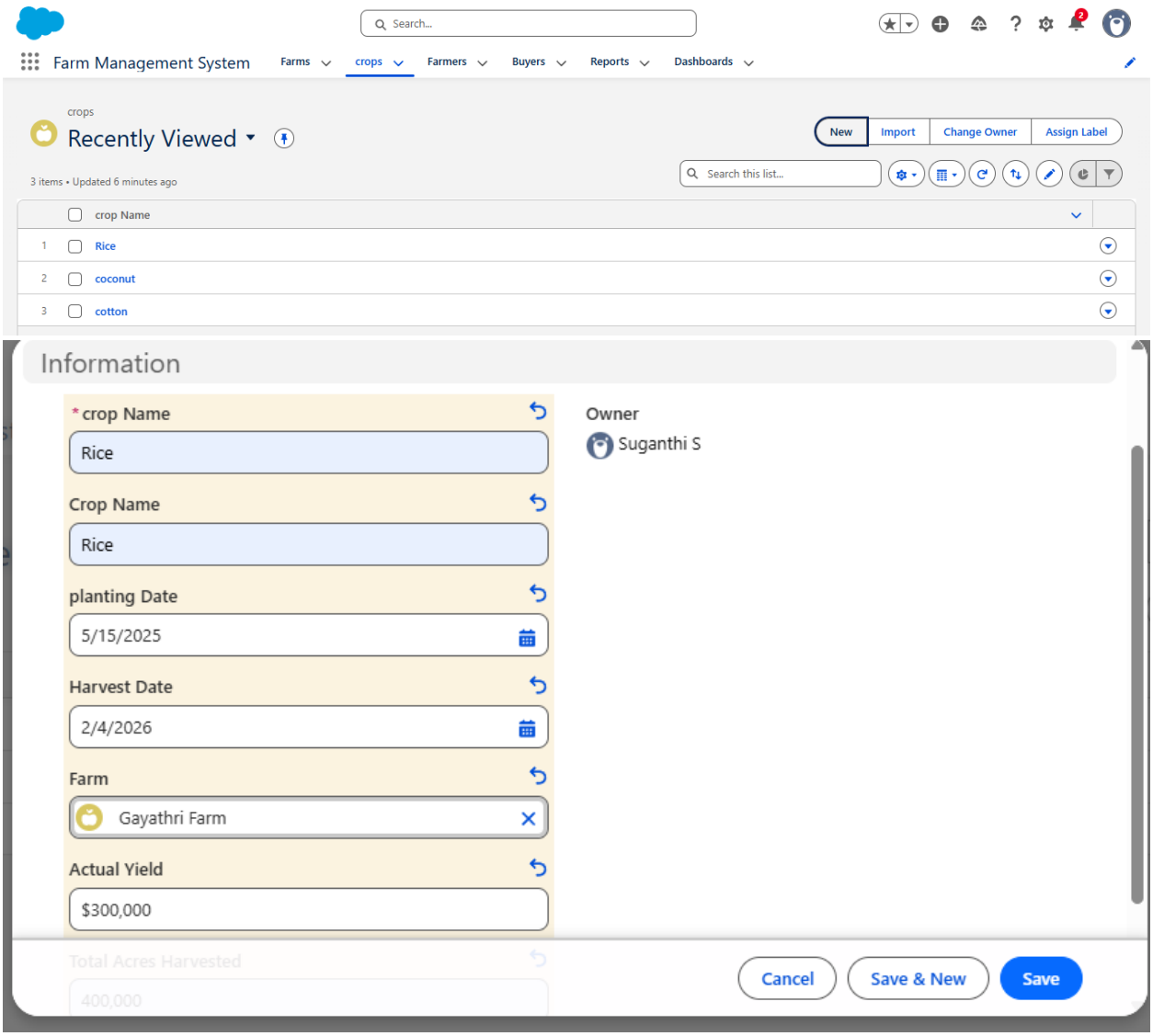
This module helps farm managers track crop production and analyze agricultural performance.

Key Fields in Crop Object

- Crop Name
- Crop Type
- Planting Date
- Harvest Date
- Expected Yield
- Actual Yield
- Harvested Area
- Farm (Lookup Relationship)

Screenshots:





11.Farmer:

The Farmer module is used to manage and maintain information about farmers who are associated with the farms in the system. This module stores the personal and contact details of farmers in a structured format.

The system records information such as farmer name, contact number, address, state, district, and village. By storing farmer data digitally, the

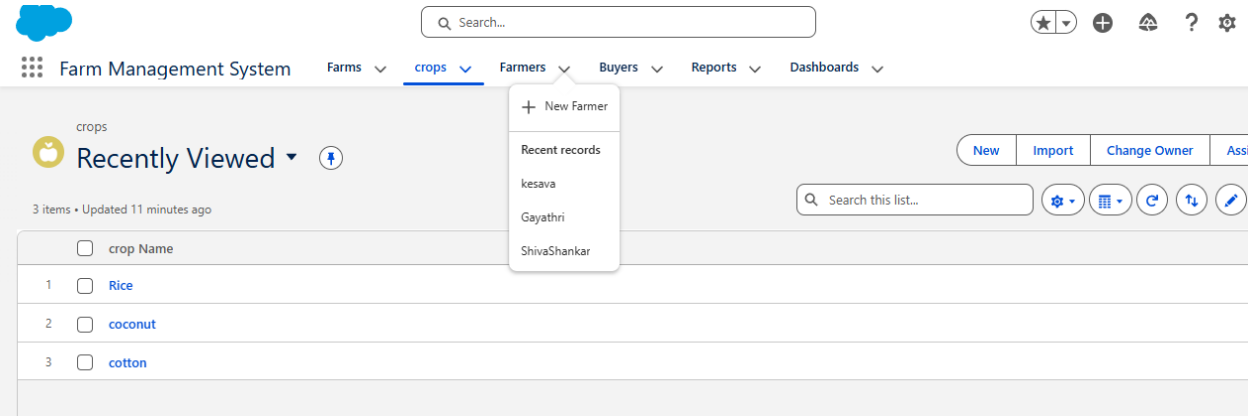
system ensures that farm administrators can easily access and manage farmer records whenever required.

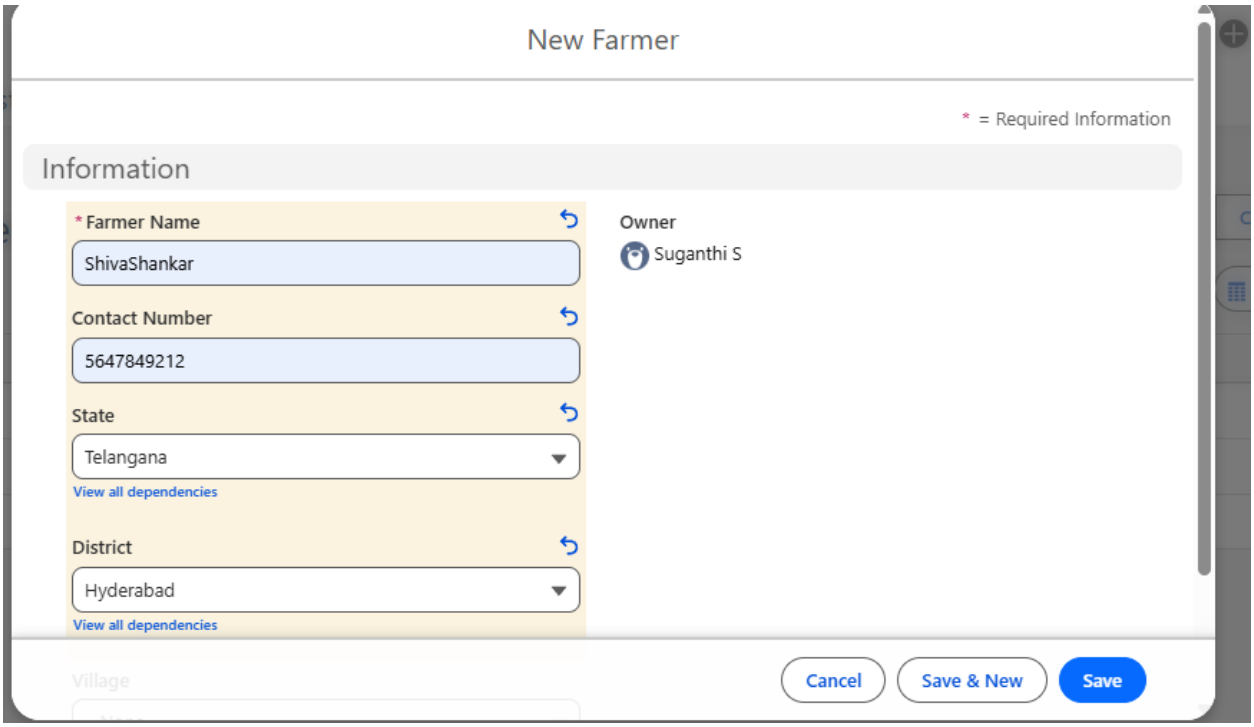
This module is linked with the Farm module using a Master-Detail relationship. This connection ensures that each farm is associated with a specific farmer, helping the system maintain proper ownership and management of details.

Main Fields in Buyer Module

- Buyer Name
- Contact Number
- Email Address
- Buyer Type
- Address
- City
- State

Screenshots:





12.Buyer:

The Buyer module manages information related to buyers who purchase agricultural products from farms. This module stores buyer details and helps track potential customers who are interested in buying crops.

The system records information such as buyer name, contact number, email address, buyer type, and location. Maintaining buyer records helps farmers and farm managers manage crop sales more efficiently.

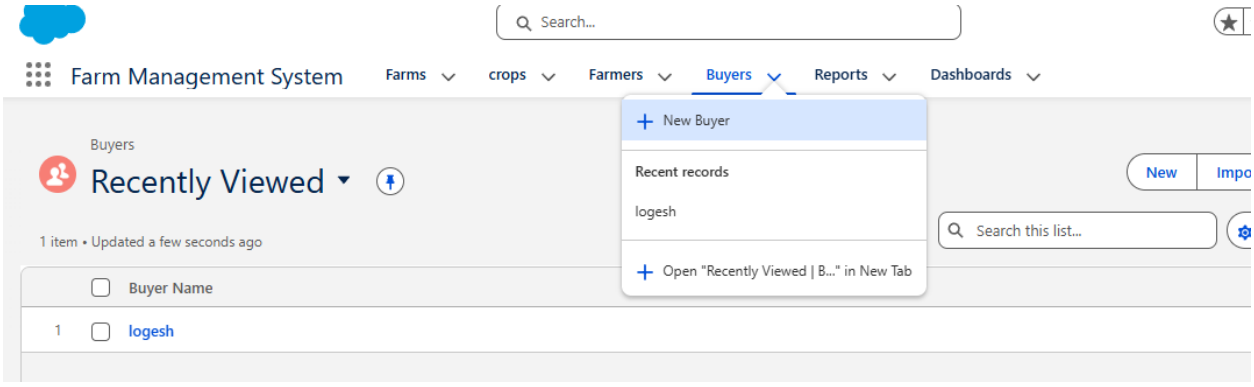
The Buyer module helps improve market connectivity by keeping track of buyers who purchase crops from farms. It also helps administrators analyze buyer demand and manage agricultural sales effectively.

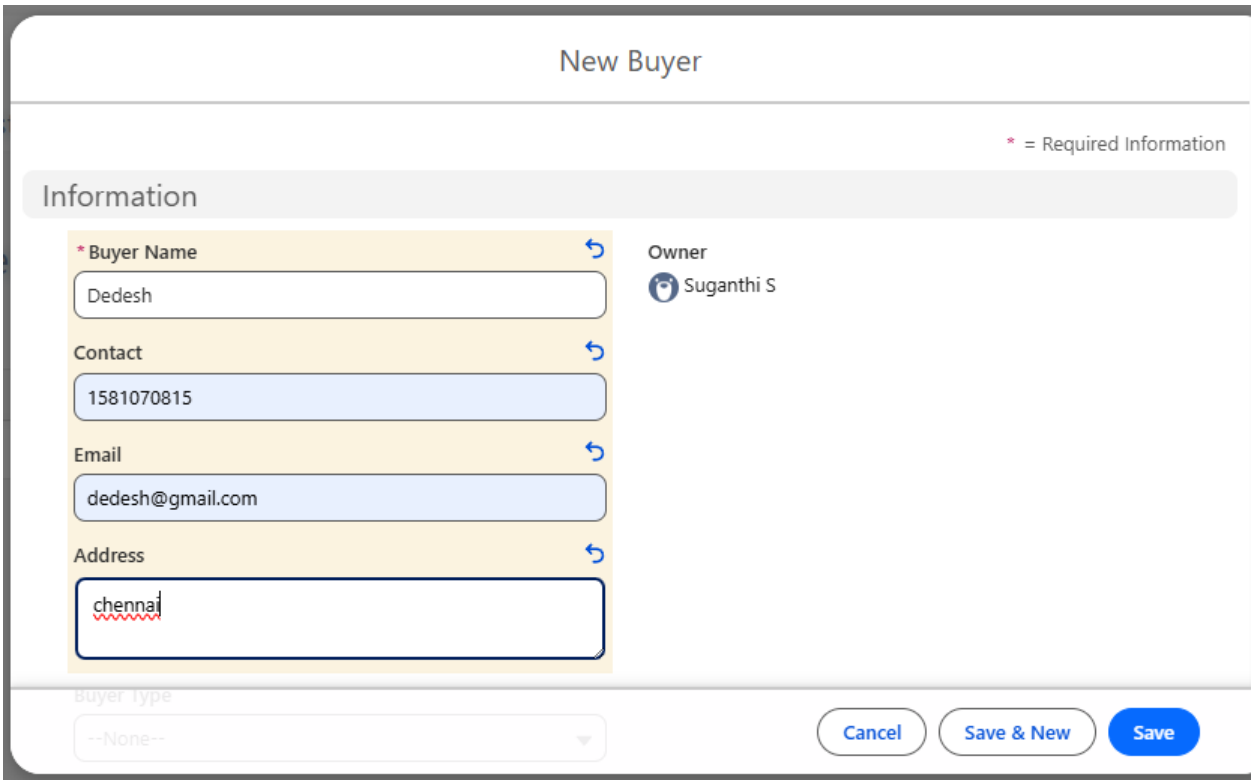
Main Fields in Buyer Module

- Buyer Name
- Contact Number
- Email Address
- Buyer Type

- Address
- City
- State

Screenshots:





13. Reports

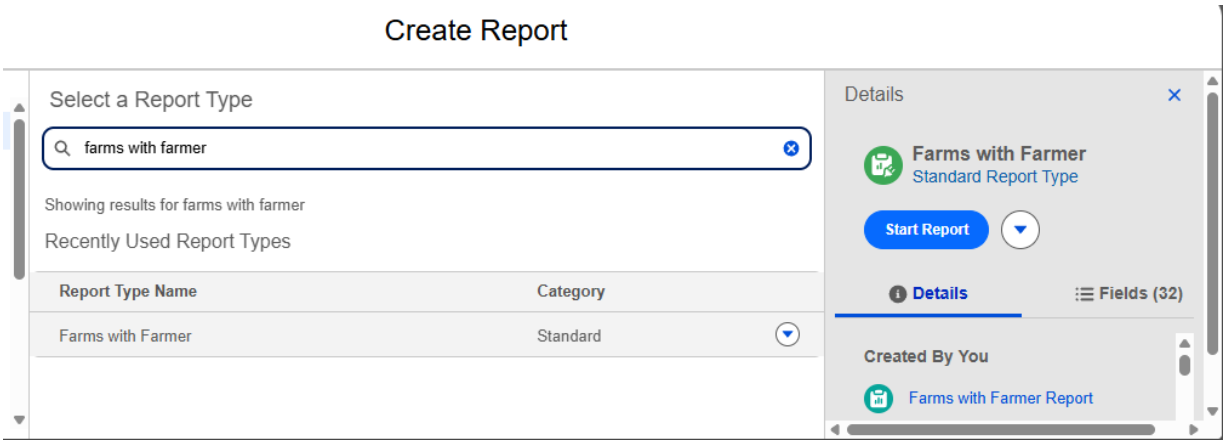
Reports are used to display farm data in a tabular format.

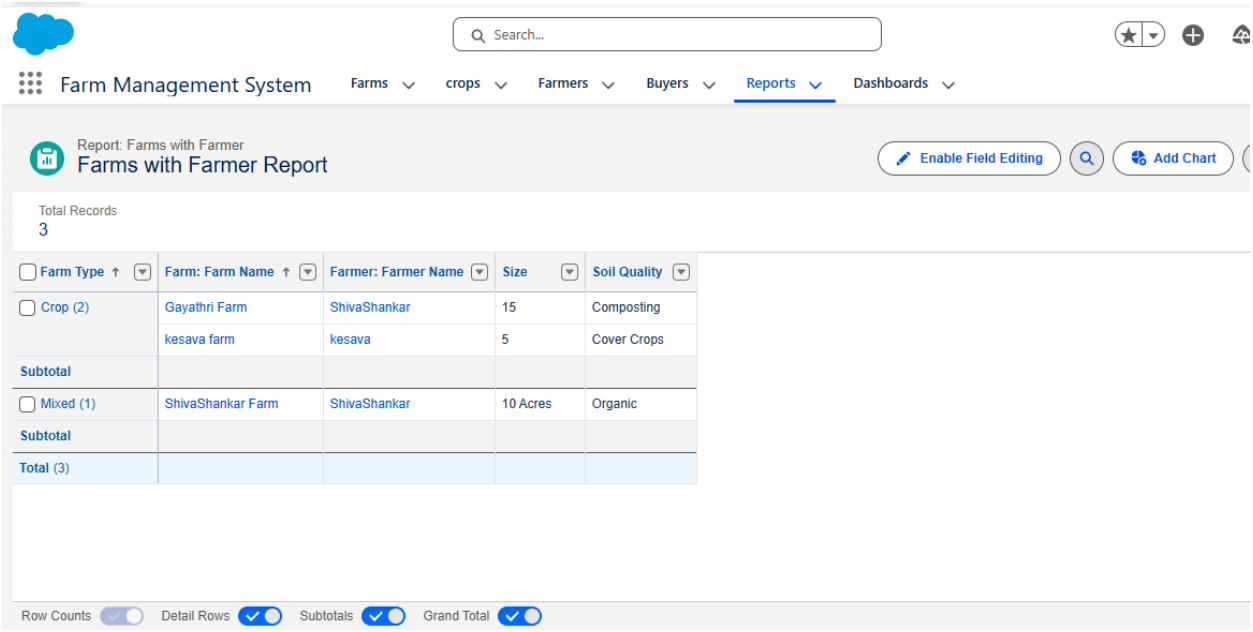
The system generates reports such as:

- Farmer with Farms Report
- Crop Production Report
- Buyer Transaction Report

Reports help farm managers analyze agricultural activities and track performance.

Screenshots:





14. Dashboards

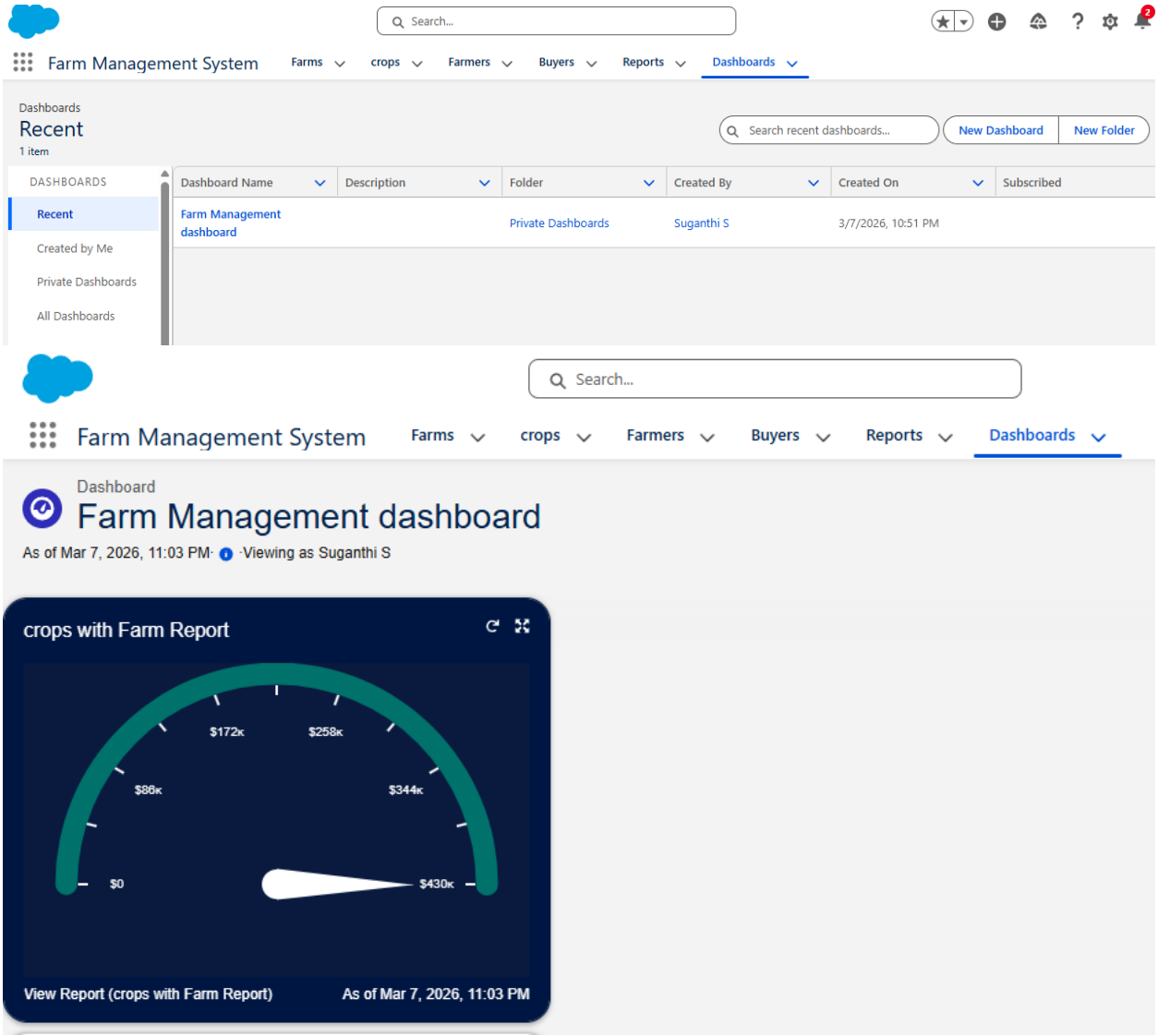
Dashboards provide visual representation of data through charts and graphs.

The dashboards display information such as:

- Crop productivity
- Farm performance
- Farmer activities
- Sales trends

This helps users quickly understand farm performance and make decisions.

Screenshots:



15.Advantages:

The Farm Management System provides several benefits:

- Centralized farm data management
- Improved accuracy of records
- Reduced paperwork
- Faster reporting and analysis
- Better decision-making

16.Limitations:

Despite its advantages, the system has some limitations:

• Requires internet connectivity
• Depends on Salesforce platform
• Limited offline functionality

17.Future Scope:

The system can be improved further by adding advanced features such as:

• Mobile application integration
• AI-based crop prediction
• Weather data integration
• IoT-based farm monitoring
• Government scheme integration

18.Conclusion:

The Farm Management System successfully digitizes agricultural data management using the Salesforce platform. The system provides efficient storage, management, and analysis of farm-related data.

By using automation, reports, and dashboards, the system improves farm productivity and helps farmers make informed decision.